

**TY.B.Sc[COMPUTER SCIENCE]**

**Sub:DATA SCIENCE**

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**Practical No. 1**

**Aim:** Practical of Data collection, Data curation and management for Unstructured data (NoSQL)

## TY.B.Sc[COMPUTER SCIENCE]

Couchdb database — rscript

Install couchdb first

**Rscript code** install.packages('sofa')

#devtools::install\_github("rop ensci/sofa")

library('sofa') #create

connection object

x<- Cushion\$n

ew()

#to check whether object

created x\$ping()

#create database ty

db\_create(x,dbna

## **TY.B.Sc[COMPUTER SCIENCE]**

```
me = 'ty')

db_list(x)

#create json doc

doc1<- '{"rollno":"01","name":"ABC","GRADE":"A"}'

doc_create(x,doc1,dbname

= "ty",docid = "a_1")

doc2<- '{"rollno":"02","name":"

PQR","GRADE":"A"}'

doc_create(x,doc2,dbname = "ty",docid = "a_2")

doc3<- '{"rollno":"03","name":"xyz","GRADE":"B","REMARK":"PASS"}'

doc_create(x,doc3,dbname = "ty",docid = "a_3")

#CHANGES FEED

db_changes(x,"ty")

#search for id > null so all

docs will display

db_query(x,dbname

= "ty",

    selector = list('_id'=list('$gt'=NULL)))$docs
```

**Sub:DATA SCIENCE**

## **TY.B.Sc[COMPUTER SCIENCE]**

#search for students with grade is A

```
db_query(x,dbname = "ty",selector = list(GRADE="A"))$docs
```

#search for students with remark =pass

```
db_query(x,dbname = "ty",selector = list(REMARK="PASS"))$docs
```

#return only certain fields where rollno>2

```
db_query(x,dbname = "ty",selector = list(rollno=list('$gt'=02)),fields=c("name","GRADE"))$docs
```

#convert the result of a query into a

data frame using jsonlite

```
library("jsonlite")
```

```
res<-db_query(x,dbname = "ty",selector =  
list('_id'=list('$gt'=NULL)),fields=c("name","rollno","GRADE","REMARK"),as="json")
```

## **TY.B.Sc[COMPUTER SCIENCE]**

```
#display json doc fromJSON(res)$docs
```

```
#doc_delete(cushion,d
```

```
bname,docid)
```

```
doc_delete(x,dbname =
```

```
"ty",docid = "a_2")
```

```
doc_get(x,dbname =
```

```
"ty",docid = "a_2")
```

```
doc2<- '{"name":"Sdrink","beer":"TEST","note":"yummy","note2":"yay"}'
```

```
doc_update(x,dbname = "ty",doc=doc2,docid="a_3",rev = "3-  
b1fb56db955b142c6efd3b3c52fe9e1b")
```

```
doc3<- '{"rollno':"0 1",
```

```
"name":"UZMA",
```

```
"GRADE":"A"}'
```

```
doc_update(x,dbname = "ty",doc=doc3,docid = "a_1",rev = "1- be7c98bddf8ea7c46f4f401ff387593d")
```

Output:

```

Console ~ /

R version 3.3.1 (2016-06-21) -- "Bug in Your Hair"
Copyright (C) 2016 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64 (64-bit)

R is free software and comes with ABSOLUTELY NO WARRANTY.
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Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

[workspace loaded from ~/.RData]

> install.packages(sofa)
Error in install.packages : object 'sofa' not found
> install.packages('sofa')
also installing the dependencies 'triebeard', 'urltools', 'httpcode', 'crul', 'jsonlite', 'mime'

There are binary versions available but the source versions are later:
      binary source needs_compilation
urltools 1.7.0 1.7.2             TRUE
crul      0.5.2 0.7.0             FALSE
jsonlite  1.5   1.6             TRUE
mime      0.5   0.6             TRUE

Binaries will be installed
trying URL 'https://cran.rstudio.com/bin/windows/contrib/3.3/triebeard_0.3.0.zip'
Content type 'application/zip' length 706397 bytes (689 KB)
downloaded 689 KB

trying URL 'https://cran.rstudio.com/bin/windows/contrib/3.3/urltools_1.7.0.zip'
Content type 'application/zip' length 824800 bytes (805 KB)
downloaded 805 KB

trying URL 'https://cran.rstudio.com/bin/windows/contrib/3.3/httpcode_0.2.0.zip'
Content type 'application/zip' length 26328 bytes (25 KB)
downloaded 25 KB

trying URL 'https://cran.rstudio.com/bin/windows/contrib/3.3/jsonlite_1.5.zip'
Content type 'application/zip' length 1159497 bytes (1.1 MB)

```

```

Console ~ /

* installing *source* package 'crul' ...
** package 'crul' successfully unpacked and MD5 sums checked
** R
** inst
** preparing package for lazy loading
** help
*** installing help indices
** building package indices
** installing vignettes
** testing if installed package can be loaded
*** arch - i386
*** arch - x64
* DONE (crul)

The downloaded source packages are in
'c:\Users\Administrator\AppData\Local\Temp\Rtmpatuuni\downloaded_packages'
> devtools::install_github("ropensci/sofa")
Error in loadNamespace(name) : there is no package called 'devtools'
> library('sofa')
warning message:
package 'sofa' was built under R version 3.3.3
> #create connection object
> x<-Cushion$new()
> #to check whether object created
> x$ping()
$couchdb
[1] "welcome"

$version
[1] "2.3.0"

$git_sha
[1] "07ea0c7"

$suuid
[1] "a42ad1ed9a458c27635222068f992a9f"

$features
$features[[1]]
[1] "pluggable-storage-engines"

$features[[2]]
[1] "scheduler"

$vendor
$vendor$name

```



```
Source
Console -/
> xyz
> #doc_delete(cushion,dbname,docid)
> doc_delete(x,dbname = "ty",docid = "a_2")
$ok
[1] TRUE

$doc
[1] "a_2"

$rev
[1] "2-82f1879cc7d73bef5574cc5cdf7c4094"

> doc_get(x,dbname = "ty",docid = "a_2")
Error: (404) - deleted
> doc2<- '{"name":"Sdrink","beer":"TEST","note":"yummy","note2":"yay"}'
> doc_update(x,dbname = "ty",doc=doc2,docid="a_3",rev = "3-b1fb56db955b142c6efd3b3c52fe9e1b"
)
$ok
[1] TRUE

$doc
[1] "a_3"

$rev
[1] "4-ee42f1c3bfc1fd646764c4d94333c038"

> doc3<- '{"rollno":"01","name":"UZMA","GRADE":"A"}'
Error: unexpected '}' in "doc3<- '{"rollno":"01",""
> doc3<- '{"rollno":"01","name":"UZMA","GRADE":"A"}'
Error: unexpected '}' in "doc3<- '{"rollno":"01",""
> doc3<- '{"rollno":"01","name":"UZMA","GRADE":"A"}'
Error: unexpected '}' in "doc3<- '{"rollno":"01",""
> doc3<- '{"rollno":"01","name":"UZMA","GRADE":"A"}'
Error: unexpected '}' in "doc3<- '{"rollno":"01",""
> doc3<- '{"rollno":"01",
+ "name":"UZMA",
+ "GRADE":"A"}'
> doc_update(x,dbname = "ty",doc=doc3,docid = "a_1",rev = "1-be7c98bddf8ea7c46f4f401ff387593
d")
$ok
[1] TRUE

$doc
[1] "a_1"

$rev
[1] "2-8e881d6a3e0fbdfd735da8ff70cff6cc"
```

```

Console -/
error: (400) - Name: ty. Only lowercase characters (a-z), digits (0-9), and any of the characters -, $, (, ), +, -, and / are allowed. Must begin with a letter.
> #create database ty
> db_create(x,dbname = 'ty')
$ok
[1] TRUE

> db_list(x)
[1] "dummy" "ty"
> #create json doc
> doc1<- '{"rollno":"01","name":"ABC","GRADE":""A"}'
> doc_create(x,doc1,dbname = "ty",docid = "a_1")
Error: lexical error: invalid char in json text.
: "01","name":"ABC","GRADE":""A}
      (right here) -----^

> #create json doc
> doc1<- '{"rollno":"01","name":"ABC","GRADE":"A"}'
> doc_create(x,doc1,dbname = "ty",docid = "a_1")
$ok
[1] TRUE

$ids
[1] "a_1"

$rev
[1] "1-be7c98bddf8ea7c46f4f401ff387593d"

> doc2<- '{"rollno":"02","name":"PQR","GRADE":"A"}'
> doc_create(x,doc2,dbname = "ty",docid = "a_2")
$ok
[1] TRUE

$ids
[1] "a_2"

$rev
[1] "1-1ddcb45704c37893389b050ddbdc440a"

> doc2<- '{"rollno":"03","name":"xyz","GRADE":"B","REMARK":"PASS"}'
> doc_create(x,doc1,dbname = "ty",docid = "a_3")
$ok
[1] TRUE

$ids
[1] "a_3"

$rev
[1] "1-be7c98bddf8ea7c46f4f401ff387593d"

```

#### Source

```

Console -/
> #CHANGES FEED
> db_changes(x,"ty")
$results
$results[[1]]
$results[[1]]$seq
[1] "1-g1AAAAF1eJzLYWBg4MhgTmEQTM4VtC5ISXLIyU90zmnILy7JAuoxJTikyF____z8rkQGpOiQFIjlkD1KXwZzImAvksSe1GcCnp6Zi04PPJAeQ5fGEbuAqasnC6PBugyNAAPoNL5xKhdaFG7nx11ByBq7x0j9gFELc19wQDAVnnt"

$results[[1]]$id
[1] "a_3"

$results[[1]]$changes
$results[[1]]$changes[[1]]
$results[[1]]$changes[[1]]$rev
[1] "1-be7c98bddf8ea7c46f4f401ff387593d"

$results[[2]]
$results[[2]]$seq
[1] "2-g1AAAAHTEJzLYWBg4MhgTmEQTM4VtC5ISXLIyU90zmnILy7JAuoxJTikyF____z8rkQGpOiQFIjlkD1KXwZzImAvksSe1GcCnp6Zi04PPJAeQ5fEoJ1wbJScmw1iSa1ICyKR6gm7PywGSDA1ACqh0PjFqF0DU7idG7QGI2vveQH0AUQtYbXyA3C-EUG"

$results[[2]]$id
[1] "a_2"

$results[[2]]$changes
$results[[2]]$changes[[1]]
$results[[2]]$changes[[1]]$rev
[1] "1-1ddcb45704c37893389b050ddbdc440a"

$results[[3]]
$results[[3]]$seq
[1] "3-g1AAAAHTEJzLYWBg4MhgTmEQTM4VtC5ISXLIyU90zmnILy7JAuoxJTikyF____z8rkQGpOiQFIjlkD1KXwZzImAvksSe1GcCnp6Zi04PPJAeQ5fEoJ1wbJScmw1iSa1ICyKR6gm7PywGSDA1ACqh0PjFqF0DU7idG7QGI2vveQH0AUQtYbEvZu1LTMVNSUwiacgB1yn2EP00sJsyTTun1J8S0BxDTQH7NagD045uy"

$results[[3]]$id
[1] "a_1"

$results[[3]]$changes
$results[[3]]$changes[[1]]

```

```
Source
Console ~/
$results[[3]]$changes
$results[[3]]$changes[[1]]
$results[[3]]$changes[[1]]$rev
[1] "1-be7c98bddf8ea7c46f4f401ff387593d"

$last_seq
[1] "3-g1AAAAI7eJyV0EEogJAQBdaQJUrSE-gvTFDbldxEp9MSJBuwy1pvoJfRm-hNsFgTIDHEbqbJNPMY8w0hzJIG1
sywkDFvMjYFgkML48nYryEQ0a-qkksDIAfbGK81lyLkikZLX01kn2vVMY4xtSrNVxh8BKkoota_zVqkuJa2HqkZAndhk
-1q6dy51oyAGv69Kh_zs172sc11uSRPkAP2umSouTckpzyyAXkpD6c8m2xCsRIY-wbjtJfTWvksmabARHun7A24cLEq"

$pending
[1] 0

> doc3<- '{"rollno":"03","name":"xyz","GRADE":"B","REMARK":"PASS"}'
> doc_create(x,doc3,dbname = "ty",docid = "a_3")
$ok
[1] TRUE

$Id
[1] "a_3"

$rev
[1] "3-b1fb56db955b142c6efd3b3c52fe9e1b"

> #CHANGES FEED
> db_changes(x,"ty")
$results
$results[[1]]
$results[[1]]$seq
[1] "3-g1AAAF1eJzLYwB94MhgTMEQTM4vTc5ISXLIyu9OzmNily7JAuoxJTikyF____z8rkQGpoiQFIJ1kd1KxwZzIn
AvksSe1GCcnp6Zi04PPJAeQSfGEbUwAqasnC6PBUGyNAApOnL5xKhdaFG7nx118yBq7xOj9gFELci9wQDCmnnv"

$results[[1]]$Id
[1] "a_3"

$results[[1]]$changes
$results[[1]]$changes[[1]]
$results[[1]]$changes[[1]]$rev
[1] "3-b1fb56db955b142c6efd3b3c52fe9e1b"
```

```
Source
Console ~/
[1] 0
> #search for id > null so all docs will display
> db_query(x,dbname = "ty",
+ selector = list('_id'=list('$gt'=NULL)))$docs
[[1]]
[[1]]$`_id`
[1] "a_1"
[[1]]$`_rev`
[1] "1-be7c98bddf8ea7c46f4f401ff387593d"
[[1]]$rollno
[1] "01"
[[1]]$name
[1] "ABC"
[[1]]$GRADE
[1] "A"
[[2]]
[[2]]$`_id`
[1] "a_2"
[[2]]$`_rev`
[1] "1-1ddcb45704c37893389b050ddbdc440a"
[[2]]$rollno
[1] "02"
[[2]]$name
[1] "PQR"
[[2]]$GRADE
[1] "A"
[[3]]
[[3]]$`_id`
[1] "a_3"
[[3]]$`_rev`
[1] "3-b1fb56db955b142c6efd3b3c52fe9e1b"
[[3]]$rollno
[1] "03"
```

```
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function Addins
Source
Console ~/
[1] "PASS"
> #search for students with grade is A
> db_query(x,dbname = "ty",selector = list(GRADE="A"))$docs
[[1]]
[[1]]$`_id`
[1] "a_1"
[[1]]$`_rev`
[1] "1-be7c98bddf8ea7c46f4f401ff387593d"
[[1]]$rollno
[1] "01"
[[1]]$name
[1] "ABC"
[[1]]$GRADE
[1] "A"
[[2]]
[[2]]$`_id`
[1] "a_2"
[[2]]$`_rev`
[1] "1-1ddcb45704c37893389b050ddbdc440a"
[[2]]$rollno
[1] "02"
[[2]]$name
[1] "PQR"
[[2]]$GRADE
[1] "A"
> #search for students with remark =pass
> db_query(x,dbname = "ty",selector = list(REMARK="PASS"))$docs
[[1]]
[[1]]$`_id`
[1] "a_3"
[[1]]$`_rev`
[1] "3-b1fb56db955b142c6efd3b3c52fe9e1b"
```

```
Source
Console
> #search for students with remark =pass
> db_query(x,dbname = "ty",selector = list(REMARK="PASS"))$docs
[[1]]
[[1]]$'_id'
[1] "a_3"

[[1]]$'_rev'
[1] "3-b1fb56db955b142c6efd3b3c52fe9e1b"

[[1]]$rollno
[1] "03"

[[1]]$name
[1] "xyz"

[[1]]$GRADE
[1] "B"

[[1]]$REMARK
[1] "PASS"

> #return only certain fields where rollno=2
> db_query(x,dbname = "ty",selector = list(rollno=list('$gt'='02')),fields=c("name","GRADE"))
$docs
[[1]]
[[1]]$name
[1] "xyz"

[[1]]$GRADE
[1] "B"

> #convert the result of a query into a data frame using jsonlite
> library("jsonlite")
Warning message:
package 'jsonlite' was built under R version 3.3.3
> res<-db_query(x,dbname = "ty",selector = list('$id'=list('$gt'=NULL)),fields=c("name","rollno","GRADE","REMARK"),as="json")
> #display json doc
> fromJSON(res)$docs
  name rollno GRADE  REMARK
1  ABC      01    A   <NA>
2  PQR      02    A   <NA>
3  xyz      03    B    PASS
> #doc_delete(cushion,dbname,docid)
> doc_delete(x,dbname = "ty",docid = "a_3")
```

```
File Edit Code View Plots Session Build Debug Profile Tools Help
Source
Console
> #doc_delete(cushion,dbname,docid)
> doc_delete(x,dbname = "ty",docid = "a_2")
$ok
[1] TRUE

$id
[1] "a_2"

$rev
[1] "2-82f1879cc7d73bef5574cc5cdf7c4094"

> doc_get(x,dbname = "ty",docid = "a_2")
Error: (404) - deleted
> doc2<- '{"name":"Sdrink","beer":"TEST","note":"yummy","note2":"yay"}'
> doc_update(x,dbname = "ty",doc=doc2,docid="a_3",rev = "3-b1fb56db955b142c6efd3b3c52fe9e1b")
$ok
[1] TRUE

$id
[1] "a_3"

$rev
[1] "4-ee42f1c3bfc1fd646764c4d94333c038"

> doc3<- '{"rollno":"01","name":"UZMA","GRADE":"A"}'
Error: unexpected ' in "doc3<- '{"rollno":"01",""
> doc3<- '{"rollno":"01","name":"UZMA","GRADE":"A"}'
Error: unexpected ' in "doc3<- '{"rollno":"01",""
> doc3<- '{"rollno":"01","name":"UZMA","GRADE":"A"}'
Error: unexpected ' in "doc3<- '{"rollno":"01",""
> doc3<- '{"rollno":"01","name":"UZMA","GRADE":"A"}'
Error: unexpected ' in "doc3<- '{"rollno":"01",""
> doc3<- '{"rollno":"01",
+ "name":"UZMA",
+ "GRADE":"A"}'
> doc_update(x,dbname = "ty",doc=doc3,docid = "a_1",rev = "1-be7c98bddf8ea7c46f4f401ff387593d")
$ok
[1] TRUE

$id
[1] "a_1"

$rev
[1] "2-8e881d6a1e0fbdfd735da8ff70cff6cc"
```

**TY.B.Sc[COMPUTER SCIENCE]**

**Sub:DATA SCIENCE**

**Practical No. 2**

**Aim:** Practical of Data collection, Data curation and management for Large-scale Data system (such as MongoDB)

- MongoDB Create database
- MongoDB Drop Database
- MongoDB Create collection
- MongoDB Drop collection
- MongoDB Insert Document
- MongoDB Query Document
- MongoDB Update Document
- Delete document in MongoDB
- MongoDB Projection
- limit() and skip() method in MongoDB
- Sorting of Documents in MongoDB
- MongoDB Indexing

Starting server with mongo or mongod C:\>**mongo**

>db Test

- **Create Database in MongoDB**

Once you are in the MongoDB shell, create the database in MongoDB by typing this command:

**use database\_name**

For example: create a database “tycs”:

> use tycs switched to  
db tycs

## TY.B.Sc[COMPUTER SCIENCE]

```
MongoDB Enterprise > use tycs
switched to db tycs
MongoDB Enterprise > show dbs
admin    0.000GB
config   0.000GB
local    0.000GB
tycs     0.000GB
MongoDB Enterprise >
```

create a collection **user** and insert a document in it.

```
> db.user.insert({name: "Asif", age: 20})
```

**O/P:** WriteResult({ "nInserted" : 1 })

```
>show dbs admin
```

```
0.000GB config
```

```
0.000GB local
```

```
0.000GB
```

```
tycs    0.000GB
```

### ▪ MongoDB Drop Database

The syntax to drop a Database is:

```
>db.dropDatabase()
```

**O/P:**

```
{ "dropped" : "Testdb", "ok" : 1 }
```

```
MongoDB Enterprise > show dbs admin
```

```
0.000GB
```

```
config 0.000GB
```

```
local 0.000GB
```

```
tycs 0.000GB
```



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**O/P:**

```
MongoDB Enterprise > db.dropDatabase()
{ "dropped" : "tycs", "ok" : 1 }
MongoDB Enterprise > show dbs
admin      0.000GB
config     0.000GB
local      0.000GB
MongoDB Enterprise >
```

### ▪ Create Collection in MongoDB

Method 1: Creating the Collection in MongoDB on the fly MongoDB Enterprise >

use tycs

switched to db tycs

```
MongoDB Enterprise > db.tycs.insert({name:"Asif khan",age:21,website:"www.google.com"})
```

**O/P:**

```
WriteResult({ "nInserted" : 1 })
```

Syntax: **db.collection\_name.find()**

```
MongoDB Enterprise > db.tycs.find()
```

**o/p:**

```
{ "_id" : ObjectId("5e410808e3755b1e06a63d1d"), "name" : "Asif khan", "age" : 21, "website" : "www.google.com" }
```

```
show collections
```

```
MongoDB Enterprise > show collections
```

## TY.B.Sc[COMPUTER SCIENCE]

### O/P:

tycs user

#### ▪ Drop collection in MongoDB

SYNTAX:

```
db.collection_name.drop()
```

MongoDB Enterprise > use students switched to db  
students

MongoDB Enterprise > **show collections**

students

teachers

tycs user

MongoDB Enterprise > **db.user.drop()**

true

MongoDB Enterprise > **show collections**

students

teacher tyccs

#### MongoDB Insert Document

Syntax to insert a document into the collection:

```
db.collection_name.insert()
```

## TY.B.Sc[COMPUTER SCIENCE]

```
> db.tyics.insert(
...   {
...     name: "ASIF",
...     age: 20,
...     email: "asif@gmail.com",
...     course: [ { name: "MongoDB", duration: 7 }, { name: "Java", duration: 30 } ]
...   }
... )
```

**O/P:**

```
WriteResult({ "nInserted" : 1 })
```

**Verification:**

Syntax:

```
db.collection_name.find()
```

```
> db.tyics.find()
{ "_id" : ObjectId("5c2d37734fa204bd77e7fc1c"), "name" : "ASIF", "age" : 20,
"email" : "asif@gmail.com", "course" : [ { "name" :
"MongoDB", "duration" : 7 }, { "name" : "Java", "duration" : 30 } ] }
```

MongoDB Example: Insert Multiple Documents in collection MongoDB Enterprise

```
> var beginners=
... [
...   "studentID":1001,
...   "studentName":"Asif",
...   "age":20
... ],
... ]
```

### ▪ MongoDB Query Document using find() method

Querying all the documents in JSON format MongoDB

```
Enterprise > db.students.find().pretty()
```

## TY.B.Sc[COMPUTER SCIENCE]

```
{
  "_id" : ObjectId("5e410f3fe3755b1e06a63d1e"), "studentID" : 1001,
  "studentName" : "Asif", "age" : 20
}
```

### ▪ Query Document based on the criteria

```
> db.students.find({StudentName : "Asif"}).pretty()
```

```
{
  "_id" : ObjectId("5c281c90c23e08d1515fd9cc"), "StudentId" :
  1001,
  "StudentName" : "Asif", "age" : 20
}
```

### ▪ Updating Document using update() method

#### Syntax:

```
db.collection_name.update(criteria, update_data)
```

```
> use tycs
```

```
switched to db tycs
```

```
> show collections
```

```
beginnersbook students
```

```
tycs
```

```
> db.createCollection("got")
```

```
{ "ok" : 1 }
```

```
> var abc = [
```

```
... {
```

```
...   "_id" : ObjectId("59bd2e73ce524b733f14dd65"),
```

```
...   "name" : "Asif",
```

```
...   "age" : 20
```

```
... },
```

```
... ];
```

```
> db.got.find().pretty()
{
  "_id" : ObjectId("59bd2e73ce524b733f14dd65"), "name" : "steve",
  "age" : 20
}
```

**To update multiple documents with the update() method:**

```
db.got.update({"name":"Jon Snow"},
{$set:{"name":"Kit Harington"}},{multi:true})
```

Updating Document using save() method

**Syntax:**

```
db.collection_name.save( {_id:ObjectId(), new_document} )
```

To get the \_id of a document, you can either type this command:

```
db.got.find().pretty()
```

```
> db.got.find({"name": "Asif"}).pretty()
{
  "_id" : ObjectId("59bd2e73ce524b733f14dd65"), "name" : "Asif",
  "age" : 20
}
> db.got.find().pretty()
{
  "_id" : ObjectId("59bd2e73ce524b733f14dd65"), "name" : "Steve",
  "age" : 20
}
```

▪ **MongoDB Delete Document from a Collection Syntax of remove() method:**

```
db.collection_name.remove(delete_criteria)
```

**Delete Document using remove() method**

```
> db.students.find().pretty()
{
  "_id" : ObjectId("59bcecc7668dcce02aaa6fed"),
  "StudentId" : 1001,
  "StudentName" : "Steve",
  "age" : 30
}
```

```
db.students.remove({"StudentId": 3333})
```

**Output:**

```
WriteResult({ "nRemoved" : 1 })
```

To verify whether the document is actually deleted. Type the following command:

```
db.students.find().pretty()
```

It will list all the documents of students collection.

```
> use tycs
switched to db tycs
> db.students.find().pretty()
{
  "_id" : ObjectId("5c281c90c23e08d1515fd9cc"), "StudentId" :
  1001,
  "StudentName" : "Asif", "age" : 20
}
{
  "_id" : ObjectId("5c2d38934fa204bd77e7fc1d"), "StudentId" :
  1001,
  "StudentName" : "Steve", "age" :
  30
}
```

**Remove all Documents**

```
db.collection_name.remove({})
```

▪ **MongoDB Projection**

**Syntax:**

```
db.collection_name.find({}, {field_key: 1 or 0})
```

```
> db.students.find().pretty()
{
  "_id" : ObjectId("5c281c90c23e08d1515fd9cc"), "StudentId" :
  1001,
  "StudentName" : "Steve", "age" :
  20
}
> db.students.find({}, {"_id": 0, "StudentId" : 1})
{ "StudentId" : 1001 }
{ "StudentId" : 1002 }
```

```
> db.students.find({}, {"_id": 0, "StudentName" : 0, "age" : 0})
{ "StudentId" : 1001 }
{ "StudentId" : 1002 }
```

▪ **MongoDB – limit( ) and skip( ) method**

The **limit()** method in MongoDB

**Syntax:**

```
db.collection_name.find().limit(number_of_documents)
db.studentdata.find({student_id : {$gt: 2002}}).pretty()
db.studentdata.find({student_id : {$gt: 2002}}).limit(1).pretty()
```

### MongoDB Skip() Method

```
db.studentdata.find({student_id : {$gt:2002}}).limit(1).skip(1).pretty()
```

- MongoDB sort() method

### Sorting Documents using sort() method

#### Syntax of sort() method:

```
db.collection_name.find().sort({field_key:1 or -1})
```

1 is for ascending order and -1 is for descending order. The default value is 1.

**For example:** collection studentdata contains following documents:

```
> db.studentdata.find().pretty()

{
  "_id" : ObjectId("59bf63380be1d7770c3982af"),
  "student_name" : "Steve",
  "student_id" : 1001,
  "student_age" : 1002
}
```

Let's display the student\_id of all the documents in **descending order**:

```
> db.studentdata.find({}, {"student_id": 1, _id:0}).sort({"student_id": -1})
{ "student_id" : 1001 }
{ "student_id" : 1002 }
```

To display the student\_id field of all the students in **ascending order**:

```
> db.studentdata.find({}, {"student_id": 1, _id:0}).sort({"student_id": 1})
{ "student_id" : 1001 }
{ "student_id" : 1002 }
```



```
> db.studentdata.find({}, {"student_id": 0, "_id": 0}).sort({"student_id": 1})
{ "student_name" : "Steve", "student_age" : 22 }
{ "student_name" : "Carol", "student_age" : 22 }
{ "student_name" : "Tim", "student_age" : 23 }
>
```

#### ▪ MongoDB Indexing with Example How to

##### create index in MongoDB

```
db.collection_name.createIndex({field_name: 1 or -1})
```

The value 1 is for ascending order and -1 is for **descending order**.

Let's create the index on student\_name field in **ascending order**:

```
db.studentdata.createIndex({student_name: 1})
```

##### Output:

```
{
  "createdCollectionAutomatically" : false,
  "numIndexesBefore" : 1,
  "numIndexesAfter" : 2,
  "ok" : 1
}
```

#### ▪ MongoDB – Finding the indexes in a collection

```
db.collection_name.getIndexes()
> db.studentdata.getIndexes()
[
```

```
{  
  "v" : 2,  
  "key" : {  
    "_id" : 1  
  },  
  "name" : "_id_",  
  "ns" : "test.studentdata"  
},  
]
```

### **Practical No. 3**

**Aim:** Practical of Principal Component Analysis

```
data_iris<-iris[1:4] Cov_data<-  
cov(data_iris) Eigen_data<-  
eigen(Cov_data)  
PCA_data<-princomp(data_iris,cor="False") Eigen_data$values  
PCA_data$sdev^2
```

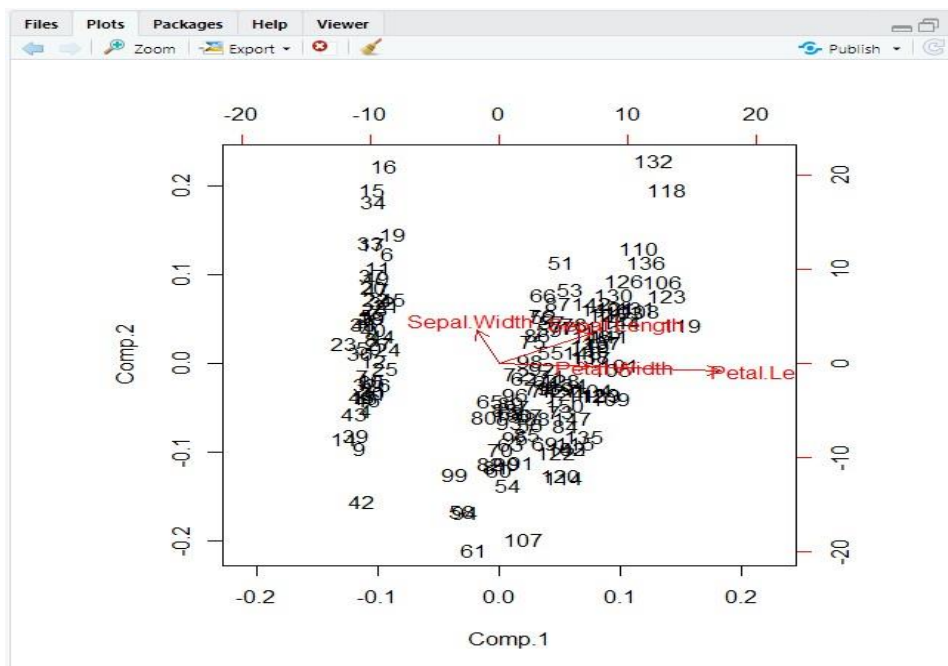
## **TY.B.Sc[COMPUTER SCIENCE]**

```
PCA_data$loadings[,1:4]
Eigen_data$vectors summary(PCA_data)
biplot(PCA_data)
screeplot(PCA_data,type="lines")
model2=PCA_data$loadings[,1]
model2_scores<-as.matrix(data_iris)%*%model2 library(class)
install.packages("e1071")
library(e1071)

mod1<-naiveBayes(iris[,1:4],iris[,5]) mod2<-
naiveBayes(model2_scores,iris[,5])
table(predict(mod1,iris[,1:4]),iris[,5])
table(predict(mod2,model2_scores),iris[,5])
```

**Output:**

```
> data_iris<-iris[1:4]
> Cov_data<-cov(data_iris)
> Eigen_data<-eigen(Cov_data)
> PCA_data<-princomp(data_iris,cor="False")
> Eigen_data$values
[1] 4.22824171 0.24267075 0.07820950 0.02383509
> PCA_data$sdev^2
      Comp.1      Comp.2      Comp.3      Comp.4
4.20005343 0.24105294 0.07768810 0.02367619
> PCA_data$loadings[,1:4]
      Comp.1      Comp.2      Comp.3      Comp.4
Sepal.Length 0.36138659 0.65658877 0.58202985 0.3154872
Sepal.Width  -0.08452251 0.73016143 -0.59791083 -0.3197231
Petal.Length 0.85667061 -0.17337266 -0.07623608 -0.4798390
Petal.Width 0.35828920 -0.07548102 -0.54583143 0.7536574
> Eigen_data$vectors
      [,1]      [,2]      [,3]      [,4]
[1,] 0.36138659 -0.65658877 -0.58202985 0.3154872
[2,] -0.08452251 -0.73016143 0.59791083 -0.3197231
[3,] 0.85667061 -0.17337266 0.07623608 -0.4798390
[4,] 0.35828920 0.07548102 0.54583143 0.7536574
> summary(PCA_data)
Importance of components:
      Comp.1      Comp.2      Comp.3      Comp.4
Standard deviation 2.0494032 0.49097143 0.27872586 0.153870700
Proportion of Variance 0.9246187 0.05306648 0.01710261 0.005212184
Cumulative Proportion 0.9246187 0.97768521 0.99478782 1.000000000
> biplot(PCA_data)
> |
```



```
naiveBayes(iris[,1:4], iris[,5])
```

60% downloaded

URL: ... s://cran.rstudio.com/bin/windows/contrib/3.6/e1071\_1.7-3.zip

```
> screplot(PCA_data,type="lines")
> model2=PCA_data$loadings[,1]
> model2_scores<-as.matrix(data_iris)%*%model2
> library(class)
> install.packages("e1071")
WARNING: Rtools is required to build R packages but is not currently installed. Please
download and install the appropriate version of Rtools before proceeding:
```

<https://cran.rstudio.com/bin/windows/Rtools/>  
trying URL 'https://cran.rstudio.com/bin/windows/contrib/3.6/e1071\_1.7-3.zip'  
Content type 'application/zip' length 1023077 bytes (999 KB)  
downloaded 999 KB

package 'e1071' successfully unpacked and MD5 sums checked

The downloaded binary packages are in  
C:\Users\asifk\AppData\Local\Temp\Rtmpgv22ko\downloaded\_packages

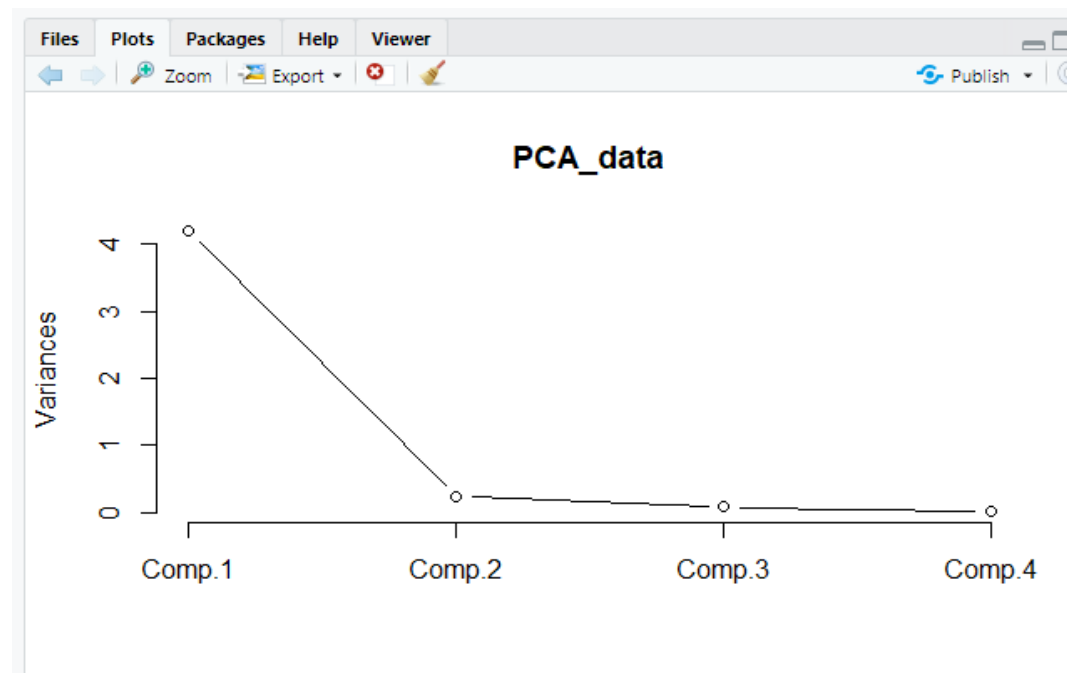
```
> library(e1071)
> mod1<-naiveBayes(iris[,1:4],iris[,5])
> mod2<-naiveBayes(model2_scores,iris[,5])
> table(predict(mod1,iris[,1:4]),iris[,5])
```

|            | setosa | versicolor | virginica |
|------------|--------|------------|-----------|
| setosa     | 50     | 0          | 0         |
| versicolor | 0      | 47         | 3         |
| virginica  | 0      | 3          | 47        |

```
> table(predict(mod2,model2_scores),iris[,5])
```

|            | setosa | versicolor | virginica |
|------------|--------|------------|-----------|
| setosa     | 50     | 0          | 0         |
| versicolor | 0      | 46         | 5         |
| virginica  | 0      | 4          | 45        |

```
> |
```



**Practical No. 4**

**Aim: Practical of Clustering**

```
"K-means Clustering " data(iris)
names(iris)
new_data<-subset(iris,select = c(-Species)) new_data
cl<-kmeans(new_data,3) cl
data<-new_data
wss<-sapply(1:15,function(k){kmeans(data,k)$tot.withinss}) wss
plot(1:15,wss,type="b",pch=19,frame=FALSE,xlab      ="Number      of
clusters K",ylab = "Total within-clusters sums of squares")
library(cluster)
clusplot(new_data,cl$cluster,color=TRUE,shade=TRUE,labels=2,lines= 0)
cl$cluster
cl$centers
"agglomerative clustering " clusters<-
hclust(dist(iris[,3:4])) plot(clusters)
clusterCut<-cutree(clusters,3)
table(clusterCut,iris$Species)
```

Output:

```

Console Terminal x Jobs x
~/
> "k-means clustering "
[1] "k-means clustering "
> data(iris)
> names(iris)
[1] "Sepal.Length" "sepal.width" "Petal.Length" "Petal.width" "species"
> new_data<-subset(iris,select = c(-Species))
> new_data
  Sepal.Length Sepal.width Petal.Length Petal.width
1           5.1          3.5           1.4          0.2
2           4.9          3.0           1.4          0.2
3           4.7          3.2           1.3          0.2
4           4.6          3.1           1.5          0.2
5           5.0          3.6           1.4          0.2
6           5.4          3.9           1.7          0.4
7           4.6          3.4           1.4          0.3
8           5.0          3.4           1.5          0.2
9           4.4          2.9           1.4          0.2
10          4.9          3.1           1.5          0.1
11          5.4          3.7           1.5          0.2
12          4.8          3.4           1.6          0.2
13          4.8          3.0           1.4          0.1
14          4.3          3.0           1.1          0.1
15          5.8          4.0           1.2          0.2
16          5.7          4.4           1.5          0.4
17          5.4          3.9           1.3          0.4
18          5.1          3.5           1.4          0.3
19          5.7          3.8           1.7          0.3
20          5.1          3.8           1.5          0.3
21          5.4          3.4           1.7          0.2
22          5.1          3.7           1.5          0.4
23          4.6          3.6           1.0          0.2
24          5.1          3.3           1.7          0.5
25          4.8          3.4           1.9          0.2
26          5.0          3.0           1.6          0.2
27          5.0          3.4           1.6          0.4
28          5.2          3.5           1.5          0.2
29          5.2          3.4           1.4          0.2
30          4.7          3.2           1.6          0.2

---
> c1<-kmeans(new_data,3)
> c1
K-means clustering with 3 clusters of sizes 96, 33, 21

Cluster means:
  Sepal.Length Sepal.width Petal.Length Petal.width
1    6.314583    2.895833    4.973958    1.7031250
2    5.175758    3.624242    1.472727    0.2727273
3    4.738095    2.904762    1.790476    0.3523810

Clustering vector:
  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20
2  3  3  3  2  2  2  2  3  3  2  2  3  3  2  2  2  2  2  2
21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40
2  2  2  2  3  3  2  2  2  3  3  2  2  2  3  2  2  2  3  2
41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60
2  3  3  2  2  3  2  3  2  2  1  1  1  1  1  1  1  3  1  1

```

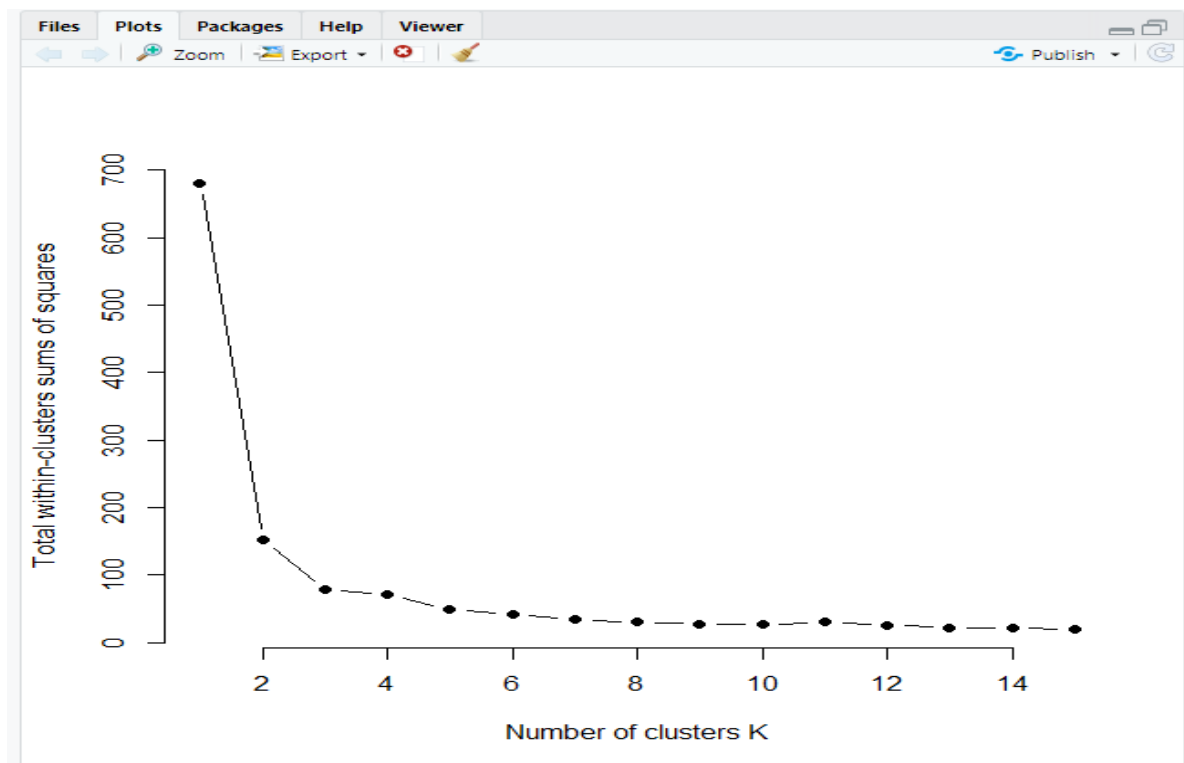
```
Source
Console Terminal x Jobs x
~/
3      4.738095      2.904762      1.790476      0.3523810

Clustering vector:
 1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20
 2  3  3  3  2  2  2  2  3  3  2  2  3  3  2  2  2  2  2  2
21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40
 2  2  2  2  3  3  2  2  2  3  3  2  2  2  3  2  2  2  3  2
41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60
 2  3  3  2  2  3  2  3  2  2  1  1  1  1  1  1  1  3  1  1
61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80
 3  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100
 1  1  1  1  1  1  1  1  1  1  1  1  1  3  1  1  1  1  3  1
101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120
 1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140
 1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
141 142 143 144 145 146 147 148 149 150
 1  1  1  1  1  1  1  1  1  1

within cluster sum of squares by cluster:
[1] 118.651875  6.432121 17.669524
      (between_SS / total_SS =  79.0 %)

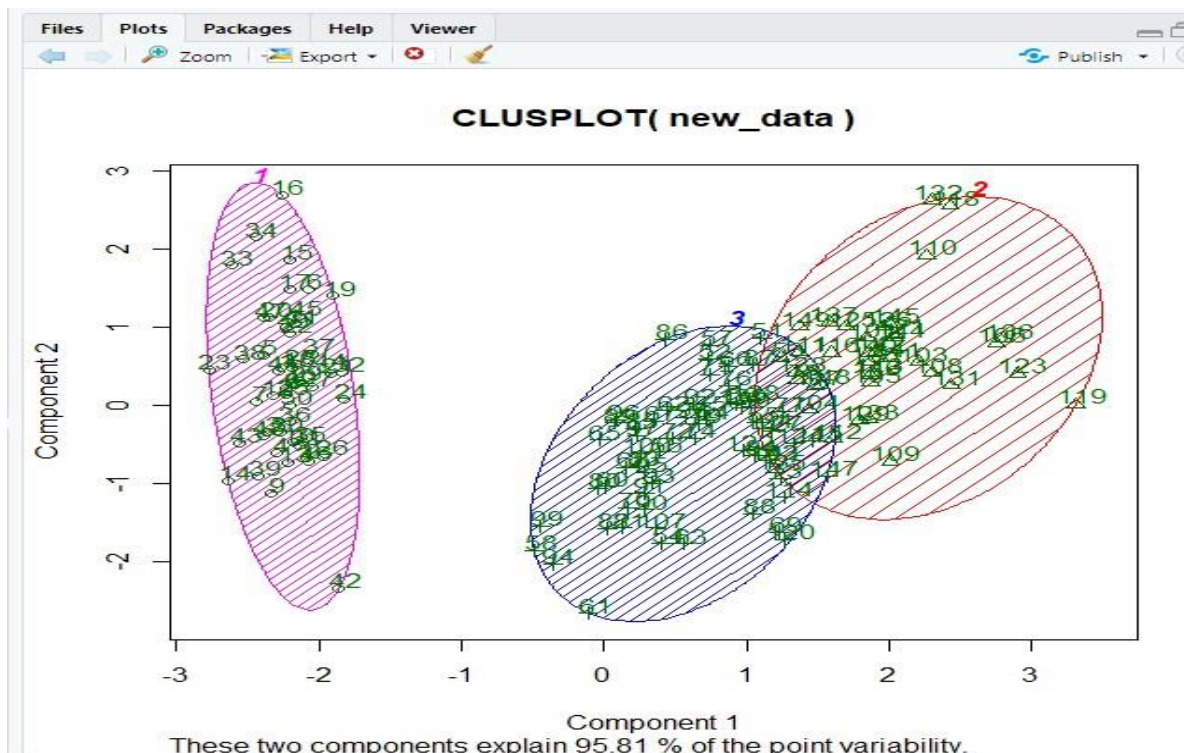
Available components:

[1] "cluster"      "centers"      "totss"        "withinss"     "tot.withinss"
[6] "betweenss"    "size"         "iter"         "ifault"       "tot.withinss"
> data<-new_data
> wss<-sapply(1:15,function(k){kmeans(data,k)$tot.withinss})
> wss
[1] 681.37060 152.34795 78.85144 71.44525 51.08942 45.49559 36.85213
[8] 46.14162 34.50212 27.59322 25.39439 23.74226 29.39379 20.70056
[15] 21.72075
> plot(1:15,wss,type="b",pch=19,frame=FALSE,xlab ="Number of clusters K",ylab = "Total within-clusters sums of squares")
> |
```



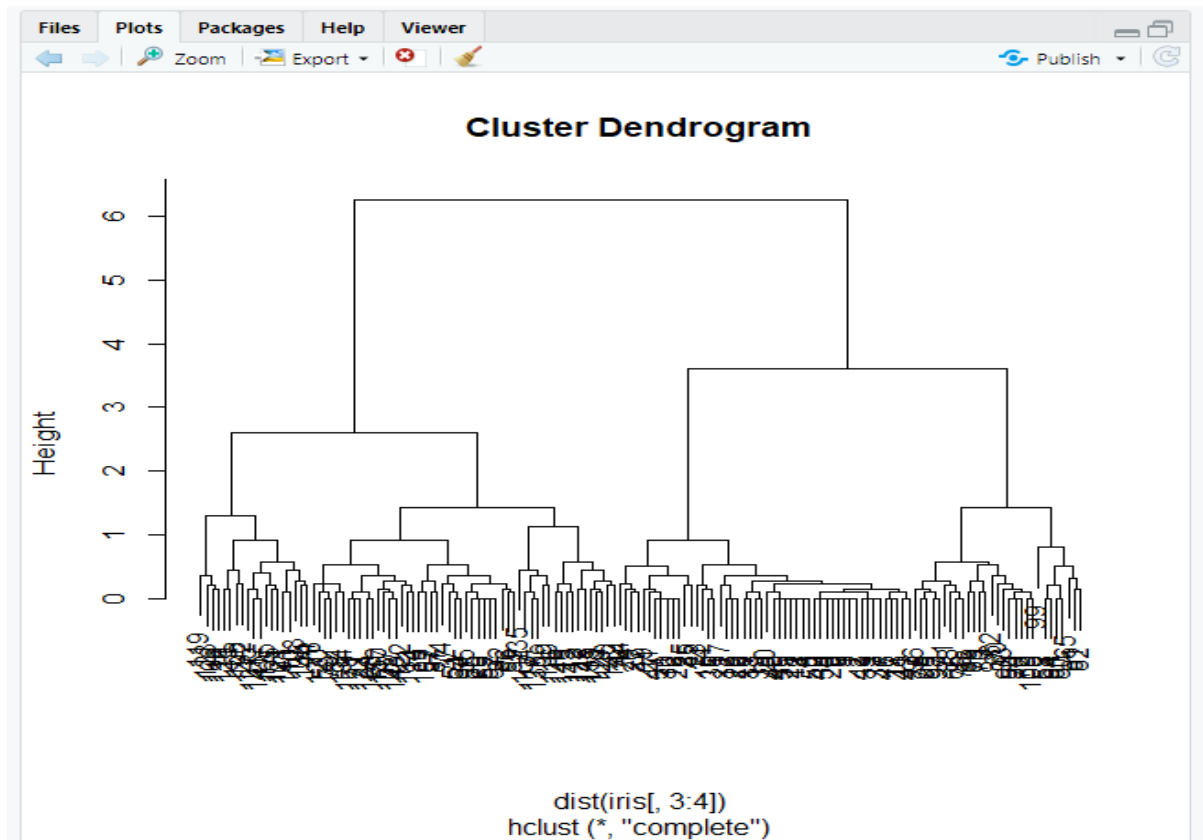


```
Source
Console Terminal x Jobs x
~/
[6] "betweenss" "size" "iter" "ifault"
> data<-new_data
> wss<-sapply(1:15,function(k){kmeans(data,k)$tot.withinss})
> wss
[1] 681.37060 152.34795 78.85144 71.44525 51.08942 45.49559 36.85213
[8] 46.14162 34.50212 27.59322 25.39439 23.74226 29.39379 20.70056
[15] 21.72075
> plot(1:15,wss,type="b",pch=19,frame=FALSE,xlab="Number of clusters K",ylab="Total within-clusters sums of squares")
> library(cluster)
> clusplot(new_data,cl$cluster,color=TRUE,shade=TRUE,labels=2,lines=0)
> cl$cluster
 1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20
 2  3  3  3  2  2  2  2  3  3  2  2  3  3  2  2  2  2  2  2
21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40
 2  2  2  2  3  3  2  2  3  3  2  2  2  3  2  2  2  2  3  2
41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60
 2  3  3  2  2  3  2  3  2  2  1  1  1  1  1  1  1  3  1  1
61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80
 3  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100
 1  1  1  1  1  1  1  1  1  1  1  1  1  3  1  1  1  1  3  1
101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120
 1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140
 1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
141 142 143 144 145 146 147 148 149 150
 1  1  1  1  1  1  1  1  1  1
> cl$centers
 Sepal.Length Sepal.Width Petal.Length Petal.Width
1 6.314583 2.895833 4.973958 1.7031250
2 5.175758 3.624242 1.472727 0.2727273
3 4.738095 2.904762 1.790476 0.3523810
> "agglomerative clustering "
[1] "agglomerative clustering "
> clusters<-hclust(dist(iris[,3:4]))
> 
```



```
> clusters<-hclust(dist(iris[,3:4]))
> plot(clusters)
> clusterCut<-cutree(clusters,3)
> table(clusterCut,iris$species)

clusterCut setosa versicolor virginica
          1      50           0         0
          2       0          21        50
          3       0          29         0
> |
```



**Practical No. 5**

**Aim:** Practical of Time-series forecasting

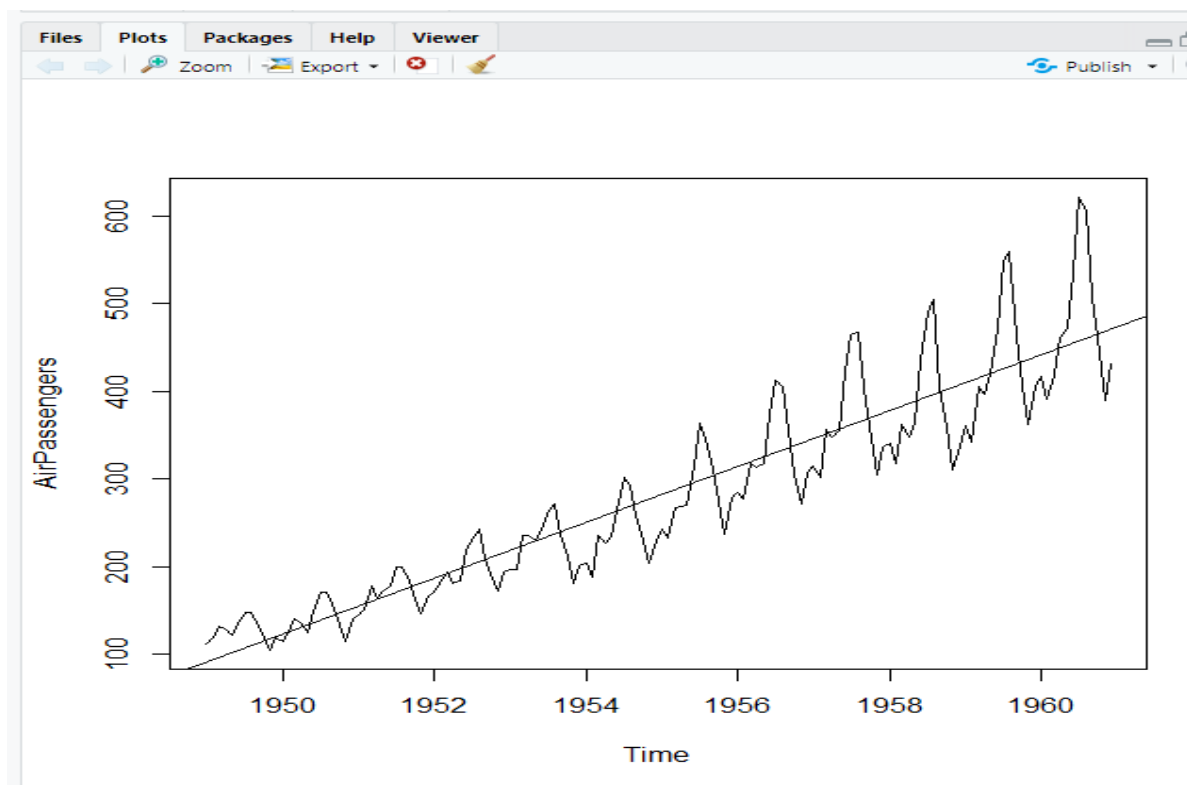
```
#consider the inbuilt data set Air Passengers data("AirPassengers")
class(AirPassengers)
start(AirPassengers)
#to know the end of time series
end(AirPassengers)
frequency(AirPassengers)
#to know the mean median etc of the dataset
summary(AirPassengers)
#to plot the time series model
plot(AirPassengers)
abline(reg=lm(AirPassengers~time(AirPassengers)))    cycle(AirPassengers)
plot(aggregate(AirPassengers,FUN=mean))
boxplot(AirPassengers~cycle(AirPassengers))
```

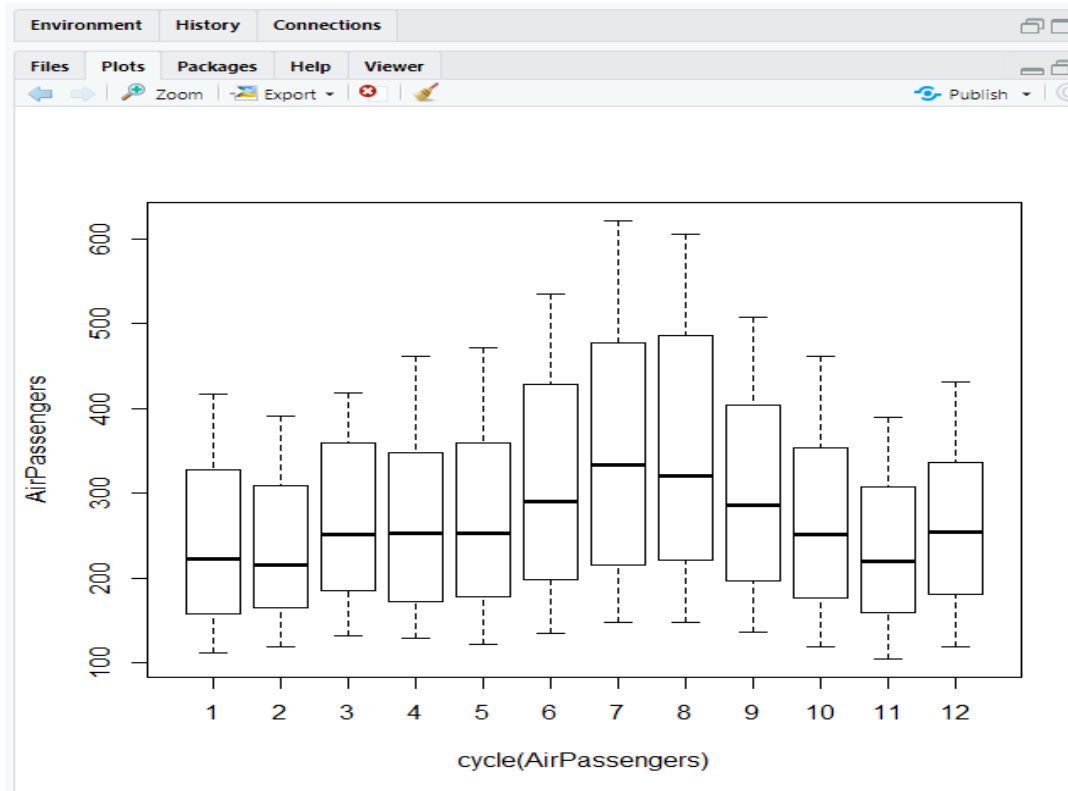
**Output:**

```

Console  Terminal x  Jobs x
~/
> data("AirPassengers")
> class(AirPassengers)
[1] "ts"
> start(AirPassengers)
[1] 1949 1
> #to know the end of time series
> end(AirPassengers)
[1] 1960 12
> frequency(AirPassengers)
[1] 12
> #to know the mean median etc of the dataset
> summary(AirPassengers)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
104.0   180.0   265.5   280.3   360.5   622.0
> #to plot the time series model
> plot(AirPassengers)
> abline(reg=lm(AirPassengers~time(AirPassengers)))
> cycle(AirPassengers)
      Jan Feb Mar  Apr May  Jun  Jul  Aug  Sep  Oct  Nov  Dec
1949   1   2   3   4   5   6   7   8   9  10  11  12
1950   1   2   3   4   5   6   7   8   9  10  11  12
1951   1   2   3   4   5   6   7   8   9  10  11  12
1952   1   2   3   4   5   6   7   8   9  10  11  12
1953   1   2   3   4   5   6   7   8   9  10  11  12
1954   1   2   3   4   5   6   7   8   9  10  11  12
1955   1   2   3   4   5   6   7   8   9  10  11  12
1956   1   2   3   4   5   6   7   8   9  10  11  12
1957   1   2   3   4   5   6   7   8   9  10  11  12
1958   1   2   3   4   5   6   7   8   9  10  11  12
1959   1   2   3   4   5   6   7   8   9  10  11  12
1960   1   2   3   4   5   6   7   8   9  10  11  12
>

```





**Practical No. 6**

**Aim:** Practical of Simple/Multiple Linear Regression

#consider some data set

height<-

c(102,117,105,141,135,115,138,144,137,100,131,119,115,121,113) weight<-

c(61,46,62,54,60,69,51,50,46,64,48,56,64,48,59)

#lm is for Linear Regression

student<-lm(weight~height) student

#to predict use predict command

predict(student,data.frame(height=199),interval="confidence") #to plot the data

plot(student)

**Output:**

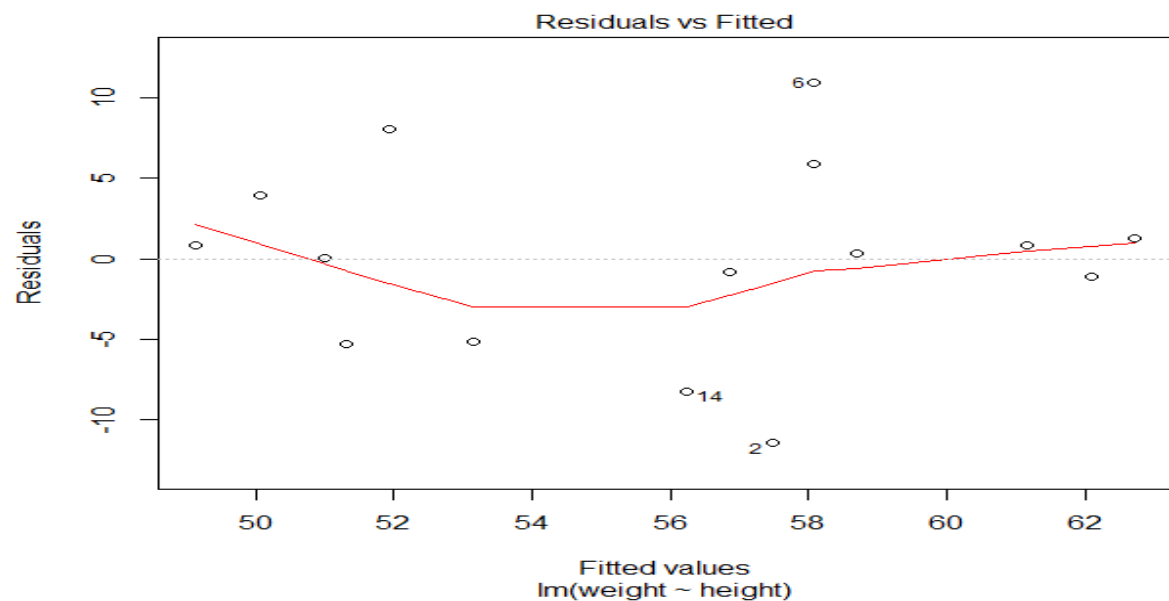
```
Console Terminal x Jobs x
~/
> #consider some data set
> height<-c(102,117,105,141,135,115,138,144,137,100,131,119,115,121,113)
> weight<-c(61,46,62,54,60,69,51,50,46,64,48,56,64,48,59)
> student<-lm(weight~height)
> student

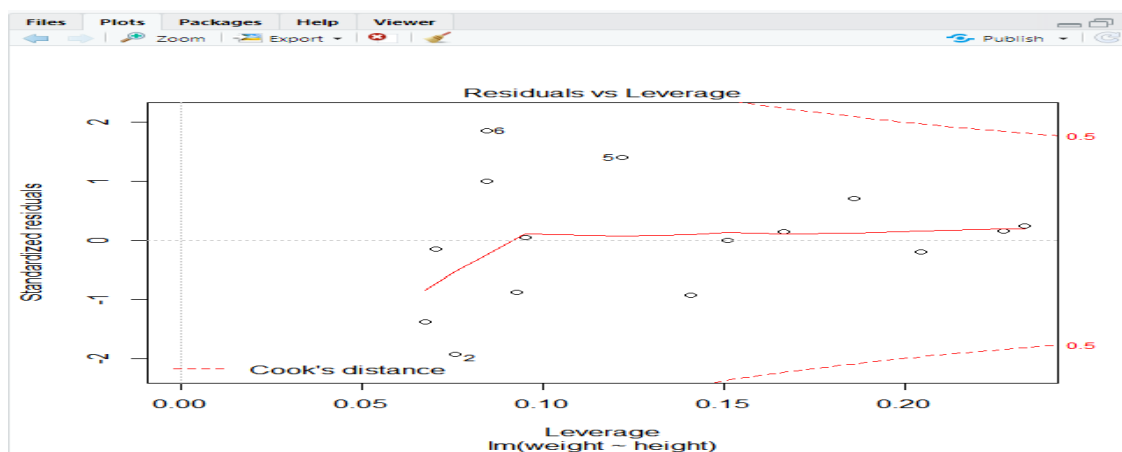
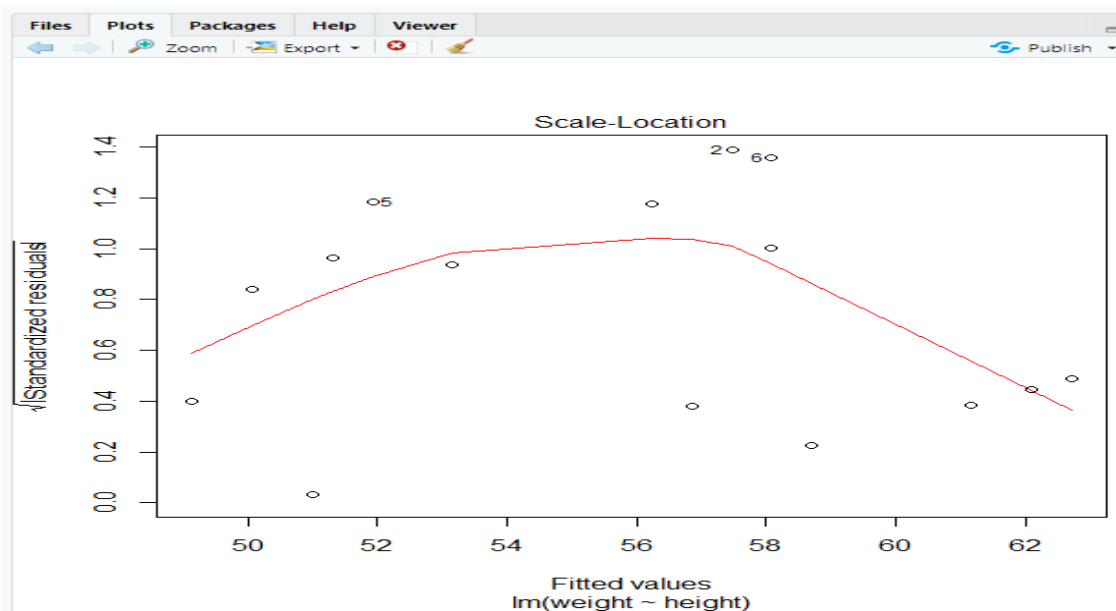
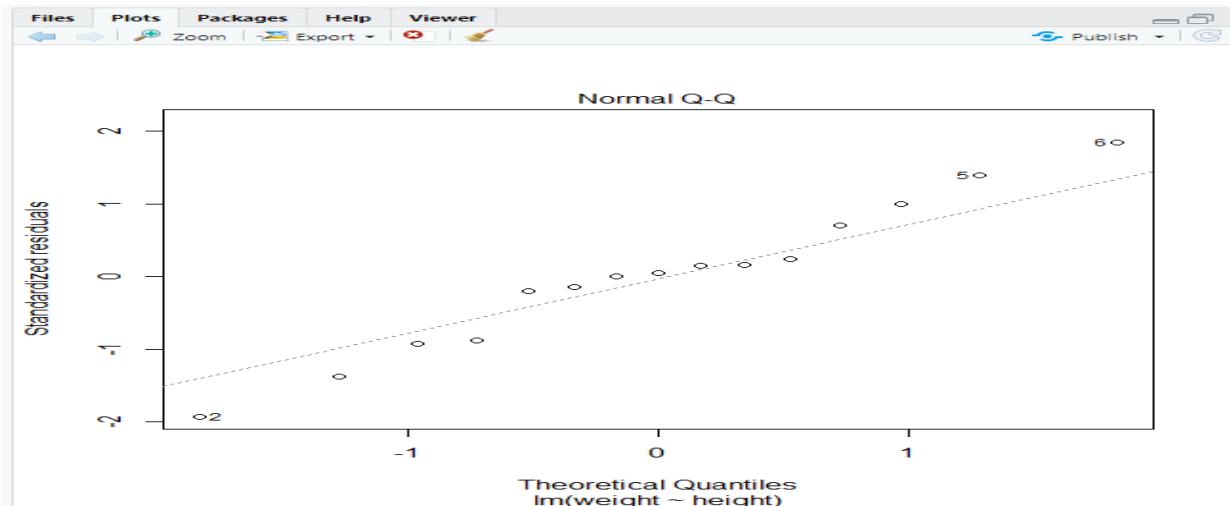
Call:
lm(formula = weight ~ height)

Coefficients:
(Intercept)      height
    93.5530      -0.3084

> predict(student,data.frame(height=199),interval="confidence")
           fit      lwr      upr
1 32.18165 13.07863 51.28468
> plot(student)
Hit <Return> to see next plot: 100
Hit <Return> to see next plot: 150
Hit <Return> to see next plot: 125
Hit <Return> to see next plot: 115
> |

Environment History Connections
Files Plots Packages Help Viewer
Zoom Export Publish
```







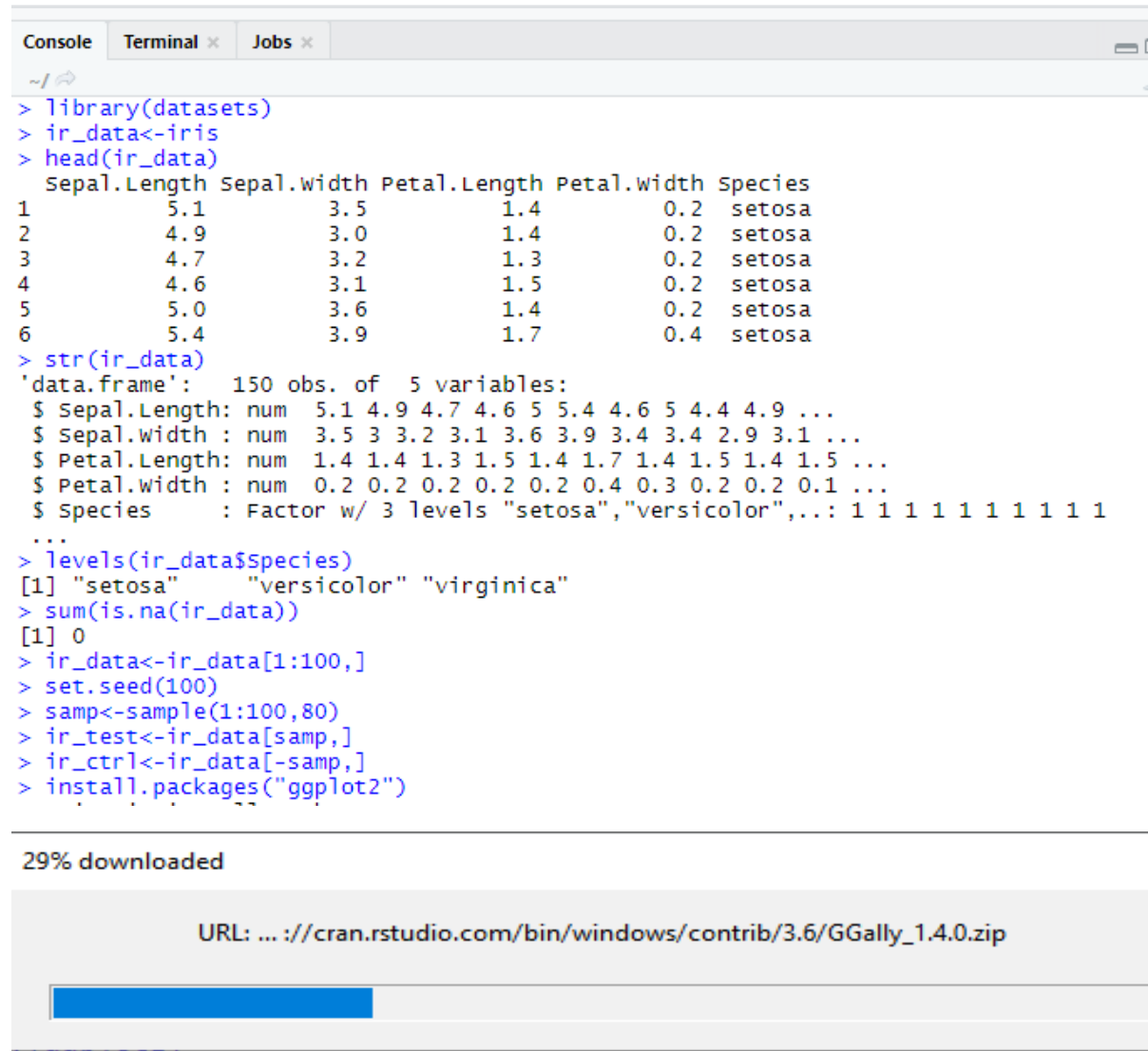
**Practical No. 7**

**Aim:** Practical of Logistics Regression

```
library(datasets) ir_data<-iris
head(ir_data) str(ir_data)
levels(ir_data$Species)
sum(is.na(ir_data)) ir_data<-
ir_data[1:100,] set.seed(100)
samp<-sample(1:100,80) ir_test<-
ir_data[samp,] ir_ctrl<-ir_data[-
samp,] install.packages("ggplot2")
library(ggplot2) library(ggplot2)
install.packages("GGally")
ggpairs(ir_test)
y<-ir_test$Species;
x<-ir_test$Sepal.Length
glfit<-glm(y~x,family='binomial')
summary(glfit)
newdata<-data.frame(x=ir_ctrl$Sepal.Length) predicted_val<-
predict(glfit,newdata,type="response")
prediction<- data.frame(ir_ctrl$Sepal.Length,ir_ctrl$Species,predicted_val)
prediction
```

qplot(prediction[,1],round(prediction[,3]),col=prediction[,2],xlab='sepal. Length',ylab='prediction using logistic Reg')

### Output:



The screenshot shows the RStudio interface. The top bar has tabs for 'Console', 'Terminal', and 'Jobs'. The 'Console' tab is active, displaying R code and its output. The code includes loading the 'datasets' library, creating an 'ir\_data' object from the 'iris' dataset, viewing the first six rows with 'head()', checking the structure with 'str()', listing the levels of the 'Species' factor, and preparing a subset of the data for modeling. Below the console, a download progress bar is visible, showing 29% completion for a file from CRAN.

```
> library(datasets)
> ir_data<-iris
> head(ir_data)
  sepal.Length sepal.width petal.Length petal.width Species
1          5.1         3.5          1.4          0.2  setosa
2          4.9         3.0          1.4          0.2  setosa
3          4.7         3.2          1.3          0.2  setosa
4          4.6         3.1          1.5          0.2  setosa
5          5.0         3.6          1.4          0.2  setosa
6          5.4         3.9          1.7          0.4  setosa
> str(ir_data)
'data.frame':   150 obs. of  5 variables:
 $ sepal.Length: num  5.1 4.9 4.7 4.6 5 5.4 4.6 5 4.4 4.9 ...
 $ sepal.width : num  3.5 3 3.2 3.1 3.6 3.9 3.4 3.4 2.9 3.1 ...
 $ petal.Length: num  1.4 1.4 1.3 1.5 1.4 1.7 1.4 1.5 1.4 1.5 ...
 $ petal.width : num  0.2 0.2 0.2 0.2 0.2 0.4 0.3 0.2 0.2 0.1 ...
 $ Species     : Factor w/ 3 levels "setosa","versicolor",...: 1 1 1 1 1 1 1 1 1 1 1
...
> levels(ir_data$Species)
[1] "setosa"      "versicolor" "virginica"
> sum(is.na(ir_data))
[1] 0
> ir_data<-ir_data[1:100,]
> set.seed(100)
> samp<-sample(1:100,80)
> ir_test<-ir_data[samp,]
> ir_ctrl<-ir_data[-samp,]
> install.packages("ggplot2")
```

29% downloaded

URL: ... ://cran.rstudio.com/bin/windows/contrib/3.6/GGally\_1.4.0.zip

**TY.B.Sc[COMPUTER SCIENCE]**

**Sub:DATA SCIENCE**

```
Console Terminal x Jobs x
~/
> y<-ir_test$Species;
> x<-ir_test$Sepal.Length
> glfit<-glm(y~x,family='binomial')
> summary(glfit)

Call:
glm(formula = y ~ x, family = "binomial")

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-2.12681  -0.51865   0.02993   0.30652   2.25044

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  -27.500      5.934  -4.634 3.59e-06 ***
x              5.112      1.109   4.611 4.01e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

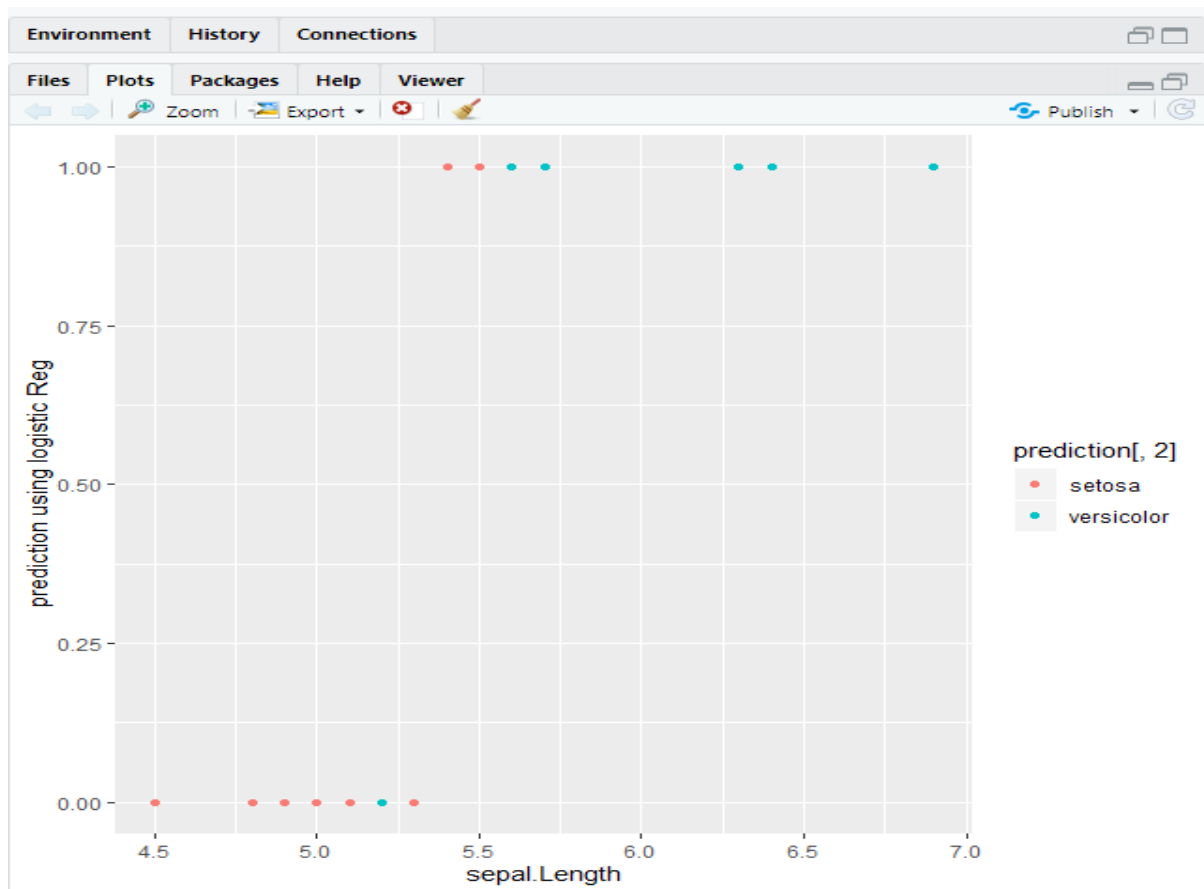
(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 110.854  on 79  degrees of freedom
Residual deviance:  48.818  on 78  degrees of freedom
AIC: 52.818

Number of Fisher Scoring iterations: 6

> newdata<-data.frame(x=ir_ctrl$Sepal.Length)
> predicted_val<-predict(glfit,newdata,type="response")
> prediction<-data.frame(ir_ctrl$Sepal.Length,ir_ctrl$Species,predicted_val)
> prediction
  ir_ctrl.Sepal.Length ir_ctrl.Species predicted_val
1                5.4      setosa      0.52665832
2                5.0      setosa      0.12584710
3                4.8      setosa      0.04923563
4                5.4      setosa      0.52665832
5                5.7      setosa      0.83759291
6                4.9      setosa      0.07948111
```

```
> newdata<-data.frame(x=ir_ctrl$sepal.Length)
> predicted_val<-predict(glmfit,newdata,type="response")
> prediction<-data.frame(ir_ctrl$sepal.Length,ir_ctrl$species,predicted_val)
> prediction
  ir_ctrl.sepal.Length ir_ctrl.species predicted_val
1                5.4         setosa    0.52665832
2                5.0         setosa    0.12584710
3                4.8         setosa    0.04923563
4                5.4         setosa    0.52665832
5                5.7         setosa    0.83759291
6                4.9         setosa    0.07948111
7                5.5         setosa    0.64975559
8                5.1         setosa    0.19357325
9                4.5         setosa    0.01104861
10               5.0         setosa    0.12584710
11               5.3         setosa    0.40023260
12               6.9        versicolor    0.99958015
13               5.7        versicolor    0.83759291
14               5.2        versicolor    0.28582944
15               5.6        versicolor    0.75569041
16               5.6        versicolor    0.75569041
17               6.3        versicolor    0.99105619
18               6.4        versicolor    0.99461661
19               5.7        versicolor    0.83759291
20               5.7        versicolor    0.83759291
> qplot(prediction[,1],round(prediction[,3]),col=prediction[,2],xlab='sepal.Length',
ylab='prediction using logistic Reg')
> |
```



**Practical No. 8**

**Aim:** Practical of Hypothesis testing

#Entering the data

```
>x=c(6.2,6.6,7.1,7.4,7.6,7.9,8,8.3,8.4,8.5,8.6,8.8,8.8,9.1,9.2,9.4,9.7,9.9,10.2,10.4,10.8,11.3,11.9)
```

#one-sample Hypothesis test

```
>t.test(x-9,alternative = "two.sided",conf.level = 0.95)
```

#Two-sample hypothesis test

```
>x=c(481,421,421,422,425,427,431,434,437,439,446,447,448,454,463,465)
```

```
>y=c(429,430,430,431,36,437,440,441,445,446,447)
```

```
>test2<-t.test(x,y,alternative = "two.sided",mu=0,var.equal = F,conf.level = 0.95)
```

```
>test2
```

**Output:**

```
Console Terminal x Jobs x
~/
> x=c(6.2,6.6,6.7,7.1,7.4,7.6,7.9,8,8.3,8.4,8.5,8.6,8.8,8.8,9.1,9.2,9.4,9.7,9.9,10.2,10.4,
10.8,11.3,11.9)
> t.test(x-9,alternative = "two.sided",conf.level = 0.95)

One sample t-test

data: x - 9
t = -0.41475, df = 22, p-value = 0.6823
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 -0.7565569  0.5043829
sample estimates:
mean of x
-0.126087

> x=c(481,421,421,422,425,427,431,434,437,439,446,447,448,454,463,465)
> y=c(429,430,430,431,36,437,440,441,445,446,447)
> test2<-t.test(x,y,alternative = "two.sided",mu=0,var.equal = F,conf.level = 0.95)
> test2

welch Two sample t-test

data: x and y
t = 1.0919, df = 10.295, p-value = 0.2998
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -41.53392 121.97710
sample estimates:
mean of x mean of y
 441.3125  401.0909

> |
```

**★ Interpretation of the result:**

The p-value (0.2998) is greater than the significance level 5%(1-0.95)

So,we conclude that the null hypothesis that the population means are equal is plausible.

**Practical No.9**

**Aim:** Practical of Analysis of Variance

```
ftest<-read.csv(file.choose(),sep="," ,header = T) var.test(ftest$time_g1,ftest$time_g2,alternative
= "two.sided")
//one way anova names(data1)
summary(data1) head(data1)
anv<-aov(formula=satindex~dept,data=data1) summary(anv)
//two way anova
data2<-read.csv(file.choose(),sep="," ,header=T) names(data2)
summary(data2) head(data2)
anv1<-aov(formula = satindex~dept+exp+dept*exp,data=data2) summary(anv1)
```



Output:

```

Console Terminal x Jobs x
~/
> ftest<-read.csv(file.choose(),sep="," ,header = T)
> var.test(ftest$time_g1,ftest$time_g2,alternative = "two.sided")

      F test to compare two variances

data:  ftest$time_g1 and ftest$time_g2
F = 1.459, num df = 13, denom df = 13, p-value = 0.5053
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
 0.4683787 4.5448881
sample estimates:
ratio of variances
      1.459016

> ftest
  time_g1 time_g2
1       85      83
2       95      85
3      105      96
4       85      94
5       90     102
6       97     100
7      104      94
8       95      95
9       88      88
10      90      92
11      94      95
12      95      94
13      86      95
14      92      90
> |

> data1<-read.csv(file.choose(),sep="," ,header=T)
> names(data1)
[1] "satindex" "dept"
> summary(data1)
      satindex      dept
Min.   :51.00   CS      :13
1st Qu.:59.00   FINANCE :12
Median :66.00   MARKETING:12
Mean    :65.59
3rd Qu.:71.00
Max.    :77.00
> head(data1)
  satindex dept
1       75 FINANCE
2       56 FINANCE
3       72 FINANCE
4       59 FINANCE
5       62 FINANCE
6       66 FINANCE
> anv<-aov(formula=satindex~dept,data=data1)
> summary(anv)
      Df Sum Sq Mean Sq F value Pr(>F)
dept    2  220.1   110.03    2.308   0.115
Residuals 34 1620.9    47.67
> |

```

```
> data2<-read.csv(file.choose(),sep="," ,header=T)
> names(data2)
[1] "satindex" "dept"      "exp"
> summary(data2)
      satindex      dept      exp
Min.   :51.00   CS      :12   gt5:18
1st Qu.:59.00   FINANCE :12   lt5:18
Median :66.00   MARKETING:12
Mean   :65.31
3rd Qu.:71.00
Max.   :77.00
> head(data2)
  satindex dept exp
1      75 FINANCE lt5
2      56 FINANCE lt5
3      72 FINANCE lt5
4      59 FINANCE lt5
5      62 FINANCE lt5
6      66 FINANCE lt5
> anv1<-aov(formula = satindex~dept+exp+dept*exp,data=data2)
> summary(anv1)
```

|           | Df | Sum Sq | Mean Sq | F value | Pr(>F) |
|-----------|----|--------|---------|---------|--------|
| dept      | 2  | 164.2  | 82.11   | 1.679   | 0.204  |
| exp       | 1  | 78.0   | 78.03   | 1.595   | 0.216  |
| dept:exp  | 2  | 20.2   | 10.11   | 0.207   | 0.814  |
| Residuals | 30 | 1467.2 | 48.91   |         |        |

```
> |
```

**Practical No.10**

**Aim:** Practical of Decision Tree

```
mydata<-data.frame(iris)
attach(mydata)
install.packages("rpart")
library(rpart)
model<- rpart(Species~Sepal.Length+Sepal.Width+Petal.Length+Petal.Width,da
ta=mydata,method="class")
plot(model) text(model,use.n=TRUE,all=TRUE,cex=0.8)
install.packages("tree")
library(tree)
model1<- tree(Species~Sepal.Length+Sepal.Width+Petal.Length+Petal.Width,data
=mydata,method="class",split="gini")
plot(model1) text(model,all=TRUE,cex=0.6)
install.packages("party") library(party)
model2<- ctree(Species~Sepal.Length+Sepal.Width+Petal.Length+Petal.Width,dat a=mydata)
plot(model2)
library(tree)
mydata<-data.frame(iris)
attach(mydata)
model<- tree(Species~Sepal.Length+Sepal.Width+Petal.Length+Petal.Width,data
=mydata,method="class",control=tree.control(nobs=150,mincut=10))
```

## **TY.B.Sc[COMPUTER SCIENCE]**

```
plot(model1)
```

```
text(model,all=TRUE,cex=0.6)
```

```
predict(model,iris)
```

```
model2<- ctree(Species~Sepal.Length+Sepal.Width+Petal.Length+Petal.Width,dat  
a=mydata,controls=ctree_control(maxdepth=2))
```

```
plot(model2)
```

**Output:**

```
Console Terminal x Jobs x
~/
> mydata<-data.frame(iris)
> attach(mydata)
> install.packages("rpart")
WARNING: Rtools is required to build R packages but is not currently installed. Please do
wnload and install the appropriate version of Rtools before proceeding:

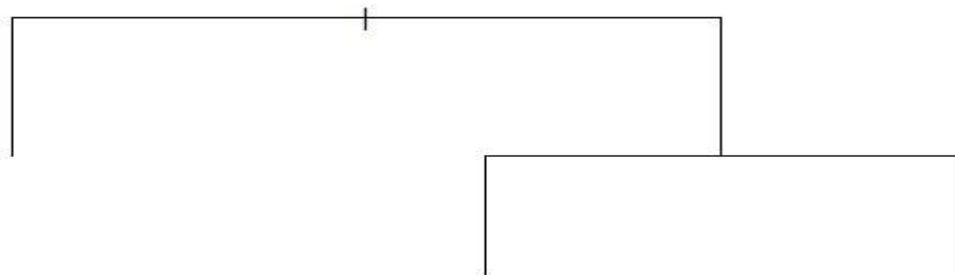
https://cran.rstudio.com/bin/windows/Rtools/
Installing package into 'C:/Users/TEMP.ITLAB.008/Documents/R/win-library/3.6'
(as 'lib' is unspecified)
trying URL 'https://cran.rstudio.com/bin/windows/contrib/3.6/rpart_4.1-15.zip'
Content type 'application/zip' length 769889 bytes (751 KB)
downloaded 751 KB

package 'rpart' successfully unpacked and MD5 sums checked

The downloaded binary packages are in
C:\Users\TEMP.ITLAB.008\AppData\Local\Temp\RtmpKAWUnZ\downloaded_packages
> library(rpart)
> model<-rpart(Species~Sepal.Length+Sepal.Width+Petal.Length+Petal.Width,data=mydata,method="class")
> plot(model)
> |
```

Files Plots Packages Help Viewer

Zoom Export Publish



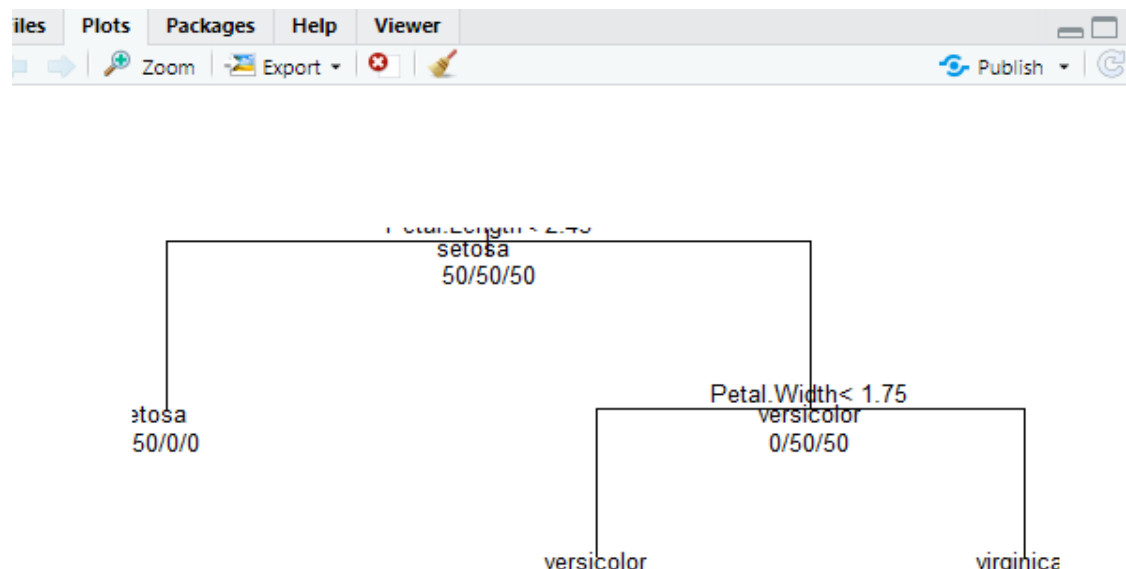
## TY.B.Sc[COMPUTER SCIENCE]

```
> text(model,use.n=TRUE,all=TRUE,cex=0.8)
> install.packages("tree")
WARNING: Rtools is required to build R packages but is not currently installed. Please
download and install the appropriate version of Rtools before proceeding:

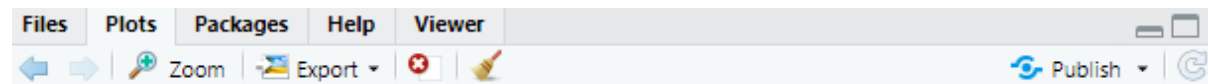
https://cran.rstudio.com/bin/windows/Rtools/
Installing package into 'C:/Users/TEMP.ITLAB.008/Documents/R/win-library/3.6'
(as 'lib' is unspecified)
trying URL 'https://cran.rstudio.com/bin/windows/contrib/3.6/tree_1.0-40.zip'
Content type 'application/zip' length 179039 bytes (174 KB)
downloaded 174 KB

package 'tree' successfully unpacked and MD5 sums checked

The downloaded binary packages are in
C:\Users\TEMP.ITLAB.008\AppData\Local\Temp\RtmpKAWUnZ\downloaded_packages
> library(tree)
> model1<-tree(Species~Sepal.Length+Sepal.Width+Petal.Length+Petal.Width,data=mydata,me
thod="class",split="gini")
> |
```



```
> model<-tree(Species~Sepal.Length+Sepal.Width+Petal.Length+Petal.Width,data=mydata,met
hod="class",control=tree.control(nobs=150,mincut=10))
> plot(model)
> model<-tree(Species~Sepal.Length+Sepal.Width+Petal.Length+Petal.Width,data=mydata,met
hod="class",control=tree.control(nobs=150,mincut=10))
> model1<-tree(Species~Sepal.Length+Sepal.Width+Petal.Length+Petal.Width,data=mydata,me
thod="class",split="gini")
> plot(model1)
> text(model,all=TRUE,cex=0.6)
> |
```



```

Console | Terminal | Jobs
~/ |
> install.packages("party")
WARNING: Rtools is required to build R packages but is not currently installed. Please
download and install the appropriate version of Rtools before proceeding:

https://cran.rstudio.com/bin/windows/Rtools/
Installing package into 'C:/Users/TEMP.ITLAB.008/Documents/R/win-library/3.6'
(as 'lib' is unspecified)
trying URL 'https://cran.rstudio.com/bin/windows/contrib/3.6/party_1.3-3.zip'
Content type 'application/zip' length 901713 bytes (880 KB)
downloaded 880 KB

package 'party' successfully unpacked and MD5 sums checked
warning in install.packages :
cannot remove prior installation of package 'party'
warning in install.packages :
problem copying C:\Users\TEMP.ITLAB.008\Documents\R\win-library\3.6\00LOCK\party\libs
\x64\party.dll to C:\Users\TEMP.ITLAB.008\Documents\R\win-library\3.6\party\libs\x64\pa
rty.dll: Permission denied
warning in install.packages :
restored 'party'

The downloaded binary packages are in
C:\Users\TEMP.ITLAB.008\AppData\Local\Temp\RtmpE1wqyz\downloaded_packages
> library(party)
Loading required package: grid
Loading required package: mvtnorm
Loading required package: modeltools
Loading required package: stats4
Loading required package: strucchange
Loading required package: zoo

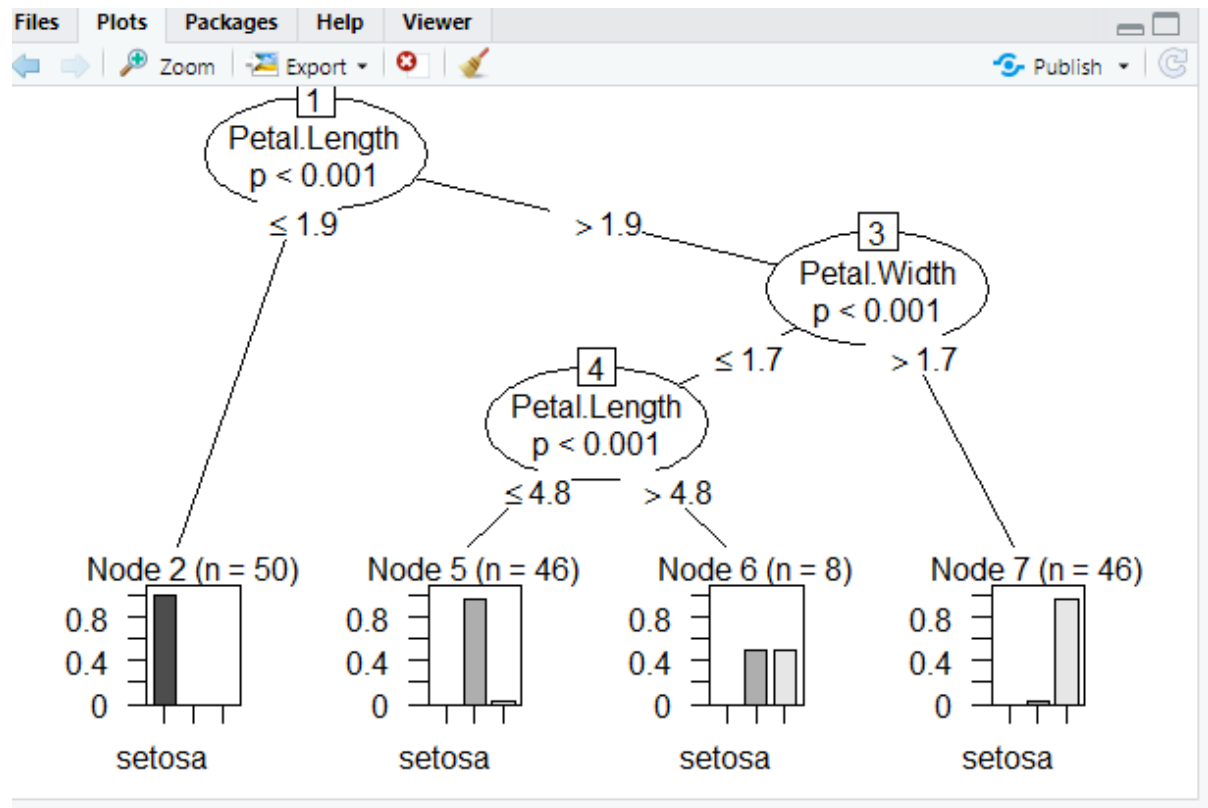
Attaching package: 'zoo'

The following objects are masked from 'package:base':

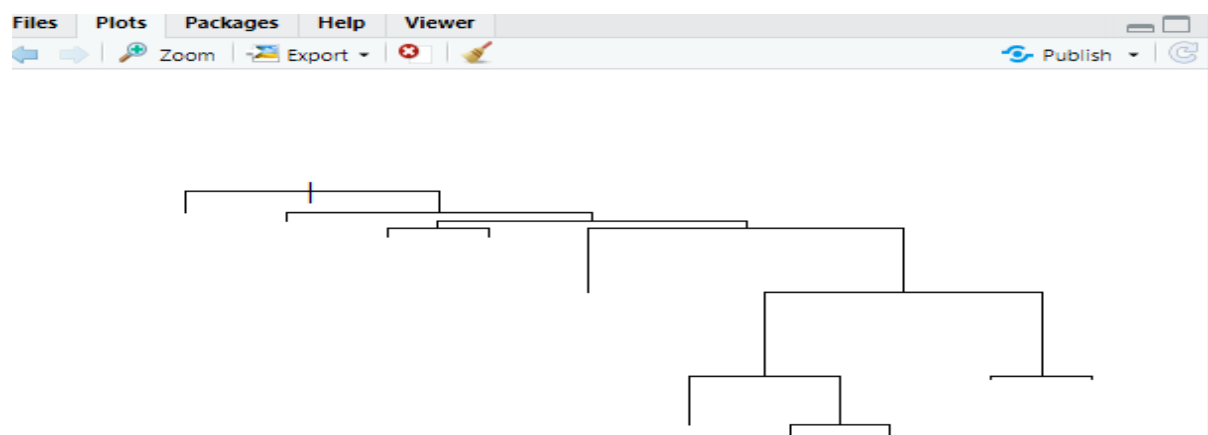
as.Date, as.Date.numeric

Loading required package: sandwich

```



```
> model2<-ctree(Species~Sepal.Length+Sepal.Width+Petal.Length+Petal.Width,data=mydata)
> plot(model2)
> library(tree)
> mydata<-data.frame(iris)
> attach(mydata)
> model<-tree(Species~Sepal.Length+Sepal.Width+Petal.Length+Petal.Width,data=mydata,met
hod="class",control=tree.control(nobs=150,mincut=10))
> plot(model1)
> |
```





```
> model2<-ctree(Species~Sepal.Length+Sepal.Width+Petal.Length+Petal.Width,data=mydata,controls=ctree_control(maxdepth=2))
> text(model,all=TRUE,cex=0.6)
> predict(model,iris)
```

|    | setosa | versicolor | virginica  |
|----|--------|------------|------------|
| 1  | 1      | 0.00000000 | 0.00000000 |
| 2  | 1      | 0.00000000 | 0.00000000 |
| 3  | 1      | 0.00000000 | 0.00000000 |
| 4  | 1      | 0.00000000 | 0.00000000 |
| 5  | 1      | 0.00000000 | 0.00000000 |
| 6  | 1      | 0.00000000 | 0.00000000 |
| 7  | 1      | 0.00000000 | 0.00000000 |
| 8  | 1      | 0.00000000 | 0.00000000 |
| 9  | 1      | 0.00000000 | 0.00000000 |
| 10 | 1      | 0.00000000 | 0.00000000 |
| 11 | 1      | 0.00000000 | 0.00000000 |
| 12 | 1      | 0.00000000 | 0.00000000 |

